PETROPHYSICS - LECTURE 11

Interactive activities student guide

Listening to microcracks: acoustic emission and the birth of a fault

Central question: how can acoustic-emission waveforms reveal where damage occurs and how the microcrack population evolves toward failure?

How to use this guide

- Work in pairs unless your instructor requests individual predictions.
- Write a prediction before moving a pick marker or changing a parameter.
- Change one control at a time and record the response.
- Distinguish measurement uncertainty from model uncertainty.
- Use reveal controls only after committing to an interpretation.

Activity	Evidence	Central question
1. Locate the source	P and S arrivals at four sensors	How do arrival times become a source location?
2. Compare b-value	Magnitude-frequency distributions	What does the event-size distribution reveal about damage?

Name: _____ Date: _____

Activity 1A - Pick arrivals and infer distance

Learning goal: convert P- and S-wave arrival picks into one source-sensor distance while recognizing that the pick itself is a measurement.

Before moving the markers, rank S1-S4 from nearest to farthest from the source using only the waveform arrival sequence.

Workflow

Step	What to do	What to notice
1	Inspect all four traces without moving the markers.	Earlier arrivals generally indicate a shorter source-sensor path.
2	Drag each P marker to the first clear onset.	The P arrival is the earliest rapid departure from background noise.
3	Drag each S marker to the later S onset.	The S phase is slower and may be less distinct in the waveform coda.
4	Select Check picks, then refine uncertain picks.	Small timing errors propagate into distance and location errors.
5	Open Convert to distance.	The P-S separation eliminates the unknown source origin time.

Equation used by the teaching model

delta t_SP = t_S - t_P -> r = delta t_SP / (1/V_S - 1/V_P)

Sensor	t_P (microseconds)	t_S (microseconds)	delta t_SP	Inferred r (mm)
S1				
S2				
S3				
S4				

Which pick was most difficult, and what feature of that waveform created the ambiguity?

Activity 1B - Combine sensors and test uncertainty

Learning goal: understand triangulation as a best-fitting intersection and separate errors caused by the velocity model from errors caused by waveform quality.

> **Predict where the source lies before selecting Solve intersection. Which sensor-distance circles should constrain it most strongly?**

From distance to location

One sensor gives a circle of possible source positions. Multiple sensors reduce the possibilities. Because measured circles rarely meet perfectly, the tool finds the position that minimizes their collective distance residuals.

Trial	Change only	Prediction	Observed location response
Baseline	Use the default velocities, attenuation and noise.		
Low V_P	Reduce assumed V_P while holding V_S fixed.		
High V_S	Increase assumed V_S while holding V_P fixed.		
High noise	Increase noise, repick P and S, then solve again.		
High attenuation	Increase attenuation and examine the distant traces.		

Interpret the controls

Control	Primary effect	Type of uncertainty
V_P and V_S	Change the conversion from travel time to distance.	Systematic model error
Noise	Makes the arrival onset ambiguous and alters picks.	Measurement error
Attenuation	Reduces waveform amplitude, especially on longer paths.	Reduced pick confidence

Why can a visually precise circle intersection still produce the wrong source location?

Scientific note

This activity uses P-S times to make the geometry visible. Laboratory AE systems often pick P arrivals at many sensors and invert simultaneously for x, y, z and origin time t_0. Clear S arrivals are not always available.

Activity 2A - Build a b-value history

Learning goal: estimate b in successive loading windows and interpret changes in the relative proportion of small and large AE events.

Predict how b should evolve as distributed microcracking becomes increasingly organized and larger events become more common.

Prior knowledge from class

log10 N(>=M) = a - bM | high b: relatively more small events | low b: relatively more large events

Guided workflow

Step	What to do	What to notice
1	Select Granite and Physical evolution.	The upper plot links stress, loading stage and AE magnitudes.
2	Choose an early, fixed-width loading window.	Only events inside the shaded window enter the distribution.
3	Set M_c where the cumulative distribution becomes approximately linear.	Events below M_c are excluded from the fit.
4	Record the current b-value.	The bottom plot stores b at the window centre with uncertainty.
5	Move the same-width window to middle and late loading.	A fair temporal comparison requires comparable analysis choices.

Granite: record three windows

Window	Start-end (%)	M_c	Events fitted	b	Interpretation
Early					
Middle					
Late					

Does a decreasing b-value prove that macroscopic failure is imminent? State one justified interpretation and one claim that would go beyond the evidence.

Activity 2B - Compare lithology and catalogue bias

Learning goal: separate physically meaningful differences between conceptual granite and sandstone scenarios from apparent b-value changes caused by catalogue completeness.

Use the same window positions and M_c strategy for both rocks. Predict which rock will retain a larger proportion of small events for longer during loading.

Granite-sandstone comparison

Stage	Granite b	Sandstone b	Microstructural interpretation
Early			
Middle			
Late			

Test the completeness trap

Select Constant b + changing completeness. Keep the analysis window fixed late in loading and vary M_c. Record what happens to the fitted line and estimated b.

M_c choice	Events fitted	Estimated b	What the distribution looks like
Too low			
Near completeness			
Conservative/high			

Synthesis

1. Which difference between granite and sandstone is consistent with contrasting microcracking mechanisms?

2. Which apparent change can be produced without any physical evolution of the generating b-value?

3. What additional evidence would you combine with b-value before inferring localization?

Final takeaway: b-value is an interpretation of a magnitude distribution. Its physical meaning depends on loading stage, lithology, window selection and catalogue completeness.